

LOWE AHP, AL, USA

WMO: 722239

Lat: 31.356N

Lon: 85.751W

Elev: 244

StdP: 14.57

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 53861

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.9	28.1	10.2	9.3	36.5	14.9	11.8	38.4	19.3	55.6	17.5	53.2	2.8	330	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	96.5	75.0	94.3	75.3	92.5	75.4	79.3	89.2	78.4	88.1	77.9	87.4	5.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	141.5	83.0	76.0	137.3	82.2	75.3	133.7	81.6	42.9	89.5	42.1	88.4	41.4	87.7	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	12.9	11.3	19.7	98.1	3.9	3.0	16.8	100.2	14.5	101.9	12.3	103.6	9.4	105.8		
			18.0	80.9	4.2	0.8	15.0	81.5	12.6	82.0	10.2	82.4	7.2	83.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.6	48.8	54.0	59.1	65.9	73.5	80.0	81.7	81.0	77.4	67.3	56.7	53.0	(6)
	DBStd	13.54	10.62	9.44	8.67	5.82	5.46	3.24	2.69	2.89	4.45	7.64	8.42	9.48	(7)
	HDD50	349	154	63	22	0	0	0	0	0	0	1	30	79	(8)
	HDD65	1850	511	320	223	58	10	0	0	0	1	66	274	387	(9)
	CDD50	6412	118	175	305	478	728	901	983	962	823	537	230	172	(10)
	CDD65	2439	10	13	41	86	273	451	518	497	373	138	24	15	(11)
	CDH74	23189	27	69	349	927	2798	4486	5091	4699	3260	1265	162	56	(12)
	CDH80	10153	0	4	53	203	1174	2206	2468	2176	1459	390	19	1	(13)
	WSAvg	3.8	4.7	4.8	5.1	4.5	3.5	3.5	2.9	2.9	2.9	3.5	3.5	4.1	(14)
	PrecAvg	57.8	5.7	4.8	5.9	4.6	3.0	5.6	6.6	4.8	4.3	3.2	4.5	3.6	(15)
(16) Precipitation	PrecMax	75.2	14.1	9.5	11.9	8.8	9.1	11.1	22.6	8.6	16.1	10.3	12.4	6.0	(16)
	PrecMin	37.2	1.9	1.6	0.4	1.0	0.0	2.6	2.7	1.8	0.7	0.5	1.5	1.5	(17)
	PrecStd	10.9	2.9	2.4	3.0	2.3	2.0	2.9	4.3	1.9	3.5	2.3	2.8	1.2	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.3	80.0	84.3	87.3	95.3	99.8	98.0	98.0	96.9	90.5	82.8	78.6	(19)
		MCWB	65.9	68.4	64.7	68.9	71.8	73.0	77.1	74.2	73.9	70.9	69.8	69.1	(20)
	2%	DB	73.4	76.0	81.4	84.7	92.0	96.5	95.4	94.8	93.6	87.3	78.2	75.4	(21)
		MCWB	65.2	65.8	65.5	67.7	71.1	74.5	76.6	76.3	73.4	70.9	66.7	67.0	(22)
	5%	DB	69.8	72.4	78.2	82.3	89.1	93.4	93.4	92.7	90.9	84.6	75.3	72.2	(23)
		MCWB	64.2	63.8	64.1	66.1	70.4	75.0	76.6	76.7	73.9	69.6	65.0	65.9	(24)
	10%	DB	66.1	69.3	74.5	79.8	86.6	91.0	91.3	90.5	88.4	81.3	72.2	68.8	(25)
		MCWB	60.7	62.2	62.4	64.9	70.0	74.9	76.9	76.1	74.2	68.0	63.5	64.4	(26)
	0.4%	WB	69.4	70.4	70.2	73.4	75.9	79.7	80.3	80.4	78.4	77.2	73.7	72.6	(27)
		MCDB	72.2	75.7	75.1	81.0	84.5	89.8	90.0	90.1	87.7	83.9	77.8	76.1	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	67.4	68.5	68.6	71.7	74.5	78.3	79.3	79.4	77.4	75.2	71.0	69.7	(29)
		MCDB	71.3	73.4	74.5	78.4	83.4	88.3	89.6	88.9	86.6	80.7	73.9	72.7	(30)
	5%	WB	65.1	66.4	67.3	69.9	73.4	77.4	78.4	78.4	76.6	73.4	68.3	67.6	(31)
		MCDB	69.0	71.0	73.5	76.5	82.5	87.2	88.6	87.6	85.6	79.8	73.3	70.2	(32)
	10%	WB	62.2	64.0	65.4	68.0	72.5	76.4	77.8	77.8	75.7	71.6	65.2	65.0	(33)
		MCDB	66.3	68.0	72.8	75.0	81.8	85.9	87.9	86.7	84.5	77.4	70.8	68.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	22.8	22.8	24.3	24.7	24.4	20.7	19.6	19.0	20.5	24.7	25.5	21.2	(35)
		MCDBR	21.9	23.2	25.3	27.4	27.5	25.2	22.2	22.3	24.3	25.6	25.2	21.6	(36)
		MCWBR	14.4	14.1	12.2	11.8	10.1	7.8	7.3	7.6	8.4	11.0	14.8	14.2	(37)
	5% WB	MCDBR	18.1	20.0	19.7	19.6	21.0	19.8	19.6	18.4	19.2	19.3	19.8	17.4	(38)
		MCWBR	13.5	14.0	11.2	11.1	8.8	7.7	7.5	7.3	7.8	9.6	13.3	13.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.325	0.334	0.358	0.383	0.435	0.472	0.517	0.504	0.432	0.373	0.347	0.331	(40)
	taud		2.520	2.496	2.431	2.364	2.253	2.209	2.095	2.127	2.309	2.448	2.468	2.519	(41)
	Ebn at Noon		287	294	293	288	272	261	249	251	266	276	276	278	(42)
	Edh at Noon		28	31	35	39	44	46	51	49	39	32	29	26	(43)
(44) All-Sky Solar Radiation	RadAvg		944	1128	1512	1853	2028	1935	1885	1753	1584	1373	1064	819	(44)
	RadStd		59	141	125	153	141	151	117	115	119	153	90	88	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-167	+299	+196

Nomenclature: See separate page